School of Future Tech

Case Study Report

on

Insurance claim system

by

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1. Introduction to the Case Study

In the healthcare and insurance industry, processing claims efficiently is critical. Manual processing is slow and error-prone. This case study presents a Health Insurance Claim Processing System built in C++ that automates the main steps of claim management.

The system allows operators to create insurance policies, submit claims, automatically validate those claims using business rules, and generate reports — all through a simple console menu. It demonstrates how structured programming and rule-based logic can solve real-world administrative problems.

2. Problem Statement / Case Background (Abstract)

Background

Insurance companies process large volumes of claims daily. Without automation, enforcing business rules consistently is difficult. A rule-based automated system can validate claims uniformly, reduce errors, and speed up processing.

Abstract

This case study implements a Health Insurance Claim Processing System in C++. The system manages policies and claims in memory, simulating a backend database. Claims are submitted against policies and then automatically validated by an engine that checks three business rules: policy activity, coverage limits, and valid claim amounts. Processed claims are displayed in a formatted report. The project uses structs, vectors, and modular functions to maintain a clean and extendable codebase.

3. Case Study Design

The system is designed as a modular, menu-driven console application with five main components:

1. Policy Management — Create policies with a unique ID, holder name, type (Gold/Silver), and coverage limit.
2. Claim Submission — Submit claims against existing policies. Claims start with a Pending status.
3. Rule-Based Validation — Automatically process all pending claims against three business rules.
4. Reporting — Display a formatted table showing every claim's ID, policy, amount, status, and reason.
5. Main Menu — A loop-driven interface allowing the user to navigate all features until exit.
    -

4. Methods & Algorithms / Technology Applied

Rule-Based Validation (validateClaim)

This function is the core of the system. It applies three rules in order:

- Rule 1 — Policy must be active. Inactive policies result in an immediate rejection.
- Rule 2 — Claim amount must not exceed the policy's maximum coverage limit.
- Rule 3 — Claim amount must be greater than zero.

If all three rules pass, the claim is marked APPROVED. Otherwise it is REJECTED with a reason.

Data Structures

- struct Policy — Stores ID, holder name, type, coverage limit, and active status.
- struct Claim — Stores claim ID, linked policy ID, amount, status, and reason.
- vector<Policy> and vector<Claim> — Dynamic arrays used as in-memory storage.
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Technology Stack

- Language: C++ (C++11 or later)
- Libraries: <iostream>, <vector>, <string>, <iomanip>
- Environment: Any C++ compiler (g++, Visual Studio, Code::Blocks, etc.)
- Storage: In-memory vectors (no file or database required)
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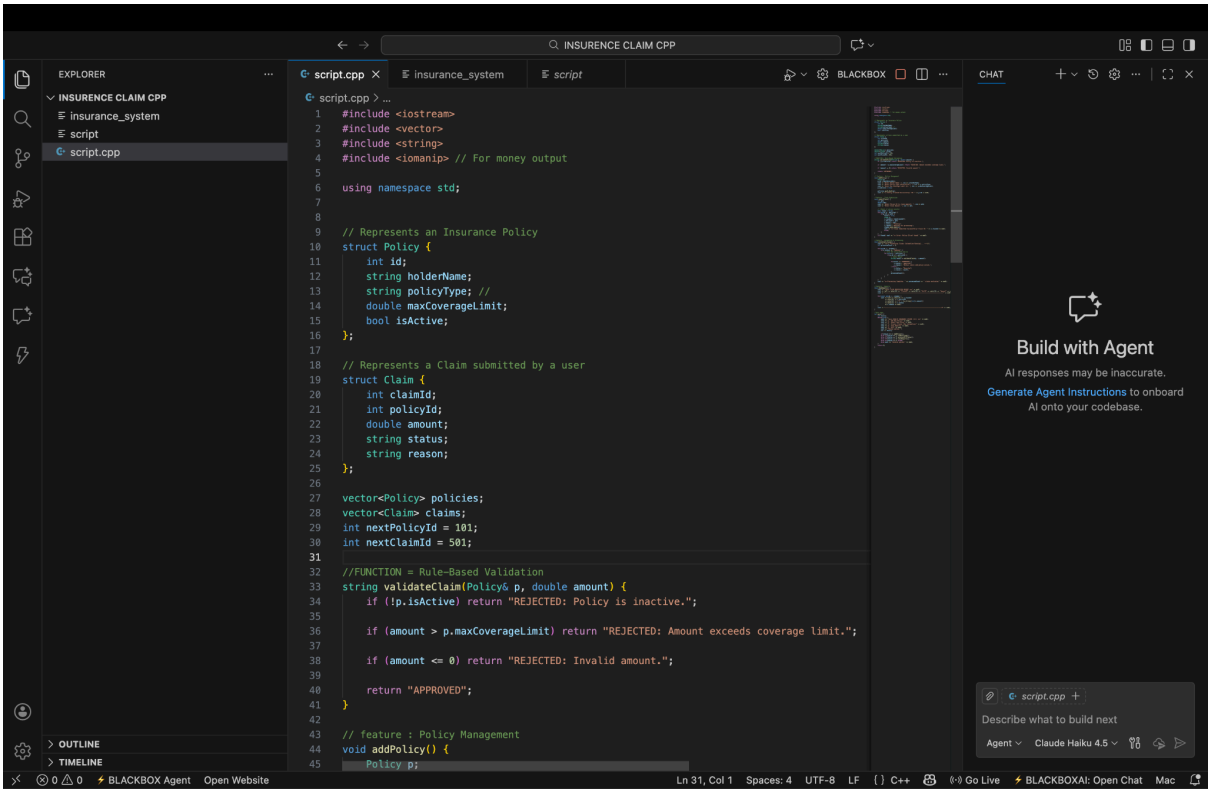
5. Problem Statement / Case Study Implementation Details and Snapshots.

- **5. Implementation Details and Snapshots**
- The entire program is in a single file (script.cpp) with the following functions:
- **addPolicy()** Asks the user for a holder name, policy type, and coverage limit. Assigns an auto-incremented ID starting at 101 and saves the policy to the global vector.
- **submitClaim()** Asks for a policy ID and claim amount. Searches for the policy and if found, creates a new claim with status Pending. Displays an error if the policy ID does not exist.
- **processAllClaims()** Loops through all pending claims, finds each claim's policy, and calls validateClaim() to apply the rules. Updates each claim's status and reason, then prints how many were evaluated.
- **showReports()** Prints a formatted table of all claims using setw() for alignment. Columns: Claim ID, Policy ID, Amount, Status, Reason.
- **main()** Runs a menu loop with five options: Add Policy, Submit Claim, Process Claims, View Rep



INSURENCE CLAIM CPP

EXPLORER

INSURENCE CLAIM CPP
- insurance_system
- script
- script.cpp

script.cpp > ...

```cpp
void addPolicy() {
    p.id = nextPolicyId++;
    cout << "Enter Holder Name: "; cin >> p.holderName;
    cout << "Enter Policy Type (Gold/Silver): "; cin >> p.policyType;
    cout << "Enter Max Coverage Limit ($): "; cin >> p.maxCoverageLimit;
    p.isActive = true;

    policies.push_back(p);
    cout << ">> Policy Created Successfully! ID: " << p.id << endl;
}

//feature : Claim Submission
void submitClaim() {
    int pId;
    double amt;
    cout << "Enter Policy ID to claim against: "; cin >> pId;
    cout << "Enter Claim Amount: "; cin >> amt;

    // (Check if policy exists)
    bool found = false;
    for(auto& p : policies) {
        if(p.id == pId) {
            found = true;
            Claim c;
            c.claimId = nextClaimId++;
            c.policyId = pId;
            c.amount = amt;
            c.status = "Pending";
            c.reason = "Waiting for processing";
            claims.push_back(c);
            cout << ">> Claim Submitted Successfully! Claim ID: " << c.claimId << endl;
            break;
        }
    }
    if(!found) cout << ">> Error: Policy ID not found." << endl;
}

//feature : Automation & Processing
void processAllClaims() {
    cout << "\n--- Processing Claims (Automation Running)... ---\n";
    int processedCount = 0;

    for(auto& c : claims) {
        if(c.status == "Pending") {
            // Find the associated policy
```

Build with Agent

AI responses may be inaccurate.

Generate Agent Instructions to onboard AI onto your codebase.

Describe what to build next

Agent · Claude Haiku 4.5

0 0 · BLACKBOX Agent · Open Website · Ln 31, Col 1 · Spaces: 4 · UTF-8 · LF · C++ · Go Live · BLACKBOXAI: Open Chat · Mac

INSURENCE CLAIM CPP

EXPLORER

INSURENCE CLAIM CPP
- insurance_system
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- script.cpp

script.cpp > ...

```cpp
void processAllClaims() {
    for(auto& c : claims) {
        if(c.status == "Pending") {
            for(auto& p : policies) {
                if(p.id == c.policyId) {
                    // Apply Rules
                    string result = validateClaim(p, c.amount);

                    if(result == "APPROVED") {
                        c.status = "Approved";
                        c.reason = "Within limits and policy active.";
                    } else {
                        c.status = "Rejected";
                        c.reason = result;
                    }
                    processedCount++;
                }
            }
        }
    }
    cout << ">> Processing Complete. " << processedCount << " claims evaluated." << endl;


//feature : Reports
void showReports() {
    cout << "\n=== CLAIM PROCESSING REPORT ===" << endl;
    cout << left << setw(10) << "ClaimID" << setw(10) << "PolID" << setw(10) << "Amount" <
    cout << "----------------------------------------------------------------------" << en

    for(const auto& c : claims) {
        cout << left << setw(10) << c.claimId
             << setw(10) << c.policyId
             << setw(10) << ("$" + to_string((int)c.amount))
             << setw(13) << c.status
             << c.reason << endl;
    }
    cout << "----------------------------------------------------------------------" << enc


//main menu
int main() {
    int choice;
    while(true) {
        cout << "\n=== HEALTH INSURANCE SYSTEM (C++) ===" << endl;
        cout << "1. Add New Policy" << endl;
```

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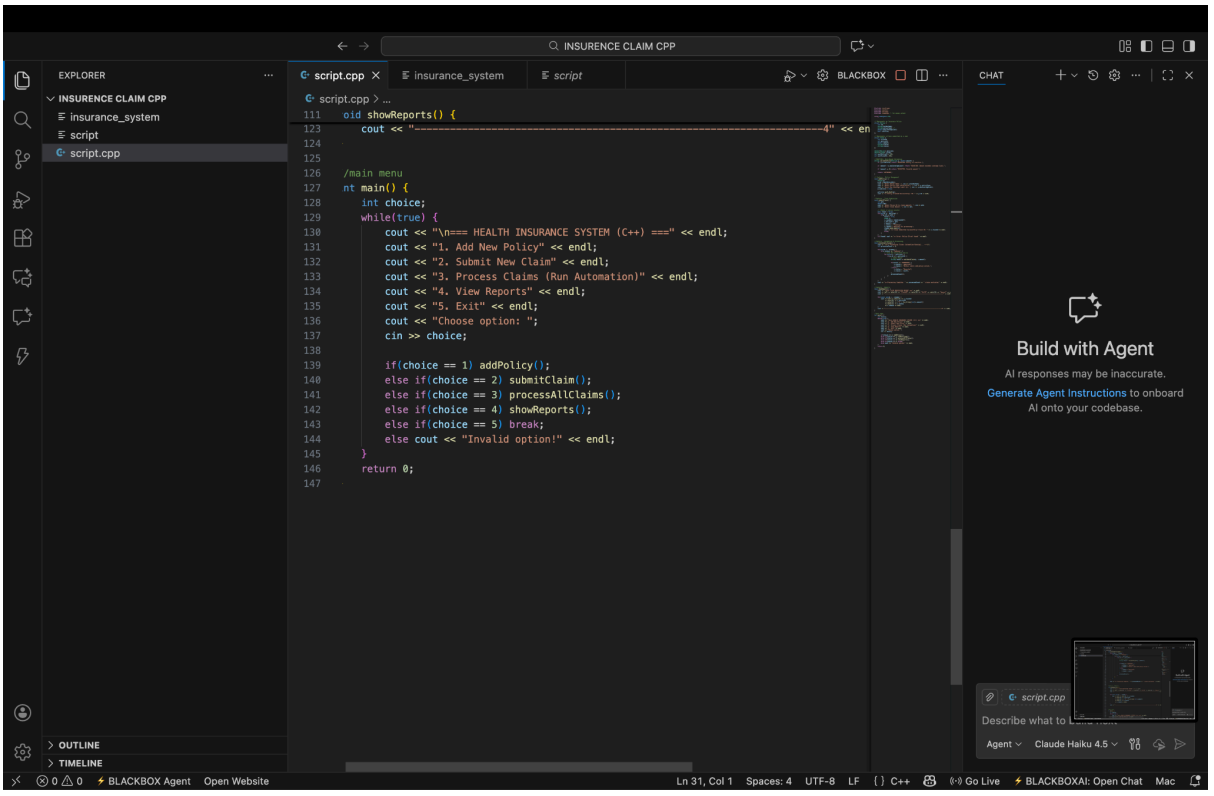

orts, and Exit.

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Sample Interaction Flow

- Create Gold policy for Alice, limit $5,000 — Policy ID: 101
- Create Silver policy for Bob, limit $2,000 — Policy ID: 102
- Submit claim of $1,500 against Policy 101 — Claim ID: 501
- Submit claim of $3,000 against Policy 102 — Claim ID: 502
- Run Process Claims — Claim 501 Approved, Claim 502 Rejected (exceeds limit)
- View Reports — table shows both claims with their status and reason

6. Results and Conclusion

Findings

- The rule-based engine correctly approves and rejects claims based on the three defined rules.
- Modular functions make the code easy to read, test, and extend.
- Auto-increment IDs ensure every record is uniquely identified.
- Formatted output using iomanip makes reports clear and readable.

Limitations

- Data is not saved to disk — all records are lost when the program closes.

- No input validation for incorrect data types.
- No support for deactivating policies or editing existing records.

Conclusion

This case study demonstrates a complete Health Insurance Claim Processing System using core C++ concepts. The project successfully integrates policy management, claim submission, automated rule-based validation, and reporting into a working console application. It can be extended with file storage, a GUI, or a database for real-world use.

7. References

- GeeksforGeeks — C++ tutorials. https://www.geeksforgeeks.org
- General domain knowledge on health insurance claim processing workflows.
- Class notes by vinaya maam